Hall–Sensor Force Model: the two normalizations of $(g_H, \hat{\mathbf{D}}_H)$

paper notation — paper = Long et al., T-Mech 2022 — dissertation = Fei Long, OSU 2016, Ch. 4

Force model

$$\mathbf{F}(\mathbf{p}, \boldsymbol{\Phi}) = g_\Phi \, \boldsymbol{\Phi}^{\mathrm{T}} \mathbf{L}(\hat{\mathbf{p}}) \, \boldsymbol{\Phi}$$

$\boldsymbol{\Phi} = [\Phi_1 \ \cdots \ \Phi_6]^{\mathrm{T}}$ is the flux vector; $\mathbf{L}(\hat{\mathbf{p}})$ is set by geometry (given); g_Φ is the force gain.

Flux from Hall voltage

$$\boldsymbol{\Phi} = \mathbf{D}_H \mathbf{V}_H, \qquad \mathbf{D}_H = \mathrm{diag}(d_{H_1}, \ldots, d_{H_6}), \qquad \mathbf{V}_H = [v_{H_1}, \ldots, v_{H_6}]^{\mathrm{T}}$$

$$\mathbf{F} = g_\Phi \, \mathbf{V}_H^{\mathrm{T}} \, \mathbf{D}_H^{\mathrm{T}} \mathbf{L}(\hat{\mathbf{p}}) \, \mathbf{D}_H \, \mathbf{V}_H$$

Unknowns to fit: g_Φ and $d_{H_1}, \ldots, d_{H_6}$ (7 in total).

Scale ambiguity

$$g_\Phi \, \mathbf{D}_H^{\mathrm{T}} \mathbf{L} \, \mathbf{D}_H = \left(g_\Phi c^{-2}\right) (c \, \mathbf{D}_H)^{\mathrm{T}} \mathbf{L} \, (c \, \mathbf{D}_H), \qquad \forall \, c \neq 0$$

$(g_\Phi, \mathbf{D}_H)$ and $(g_\Phi c^{-2}, \ c\mathbf{D}_H)$ give the same $\mathbf{F}$. So one scale is free and must be fixed before fitting.

Normalization A — Paper (six gains free)

$$\mathbf{F} = g_H \, \mathbf{V}_H^{\mathrm{T}} \hat{\mathbf{D}}_H^{\mathrm{T}} \mathbf{L}(\hat{\mathbf{p}}) \, \hat{\mathbf{D}}_H \, \mathbf{V}_H$$

$$\boxed{g_H = 4.741 \text{ pN/V}^2, \qquad \hat{\mathbf{D}}_H = \mathrm{diag}(0.444, \ 0.364, \ 0.445, \ 0.436, \ 0.353, \ 0.436)}$$

First entry is not 1: all six gains are free.

Normalization B — Dissertation (first gain = 1)

$$\hat{\mathbf{D}}_H = \mathbf{D}_H \, d_{H_1}^{-1} = \mathrm{diag}(1, \hat{d}_{H_2}, \ldots, \hat{d}_{H_6}), \qquad \underbrace{\hat{f}_H = f_\Phi \, d_{H_1}^2}_{\text{scale moved here}}$$

$$\mathbf{F} = \hat{f}_H \, \mathbf{V}_H^{\mathrm{T}} \hat{\mathbf{D}}_H^{\mathrm{T}} \mathbf{L}(\mathbf{p}, \mathbf{b}) \, \hat{\mathbf{D}}_H \, \mathbf{V}_H \qquad \text{(extra bias vector } \mathbf{b}, \text{ dissertation only)}$$

$$\boxed{\hat{f}_H = g_H \, (0.444)^2 \approx 0.93 \text{ pN/V}^2, \qquad \hat{\mathbf{D}}_H = \mathrm{diag}(1, \ 0.820, \ 1.002, \ 0.982, \ 0.795, \ 0.982)}$$

Equivalence (A and B)

$$c = 0.444 \ \left(= \hat{d}_{H_1} \text{ of the paper}\right) : \qquad \hat{\mathbf{D}}_H^{(B)} = c^{-1} \hat{\mathbf{D}}_H^{(A)}, \qquad \hat{f}_H = g_H \, c^2 \qquad \Longrightarrow \qquad \text{same } \mathbf{F}$$

Note: B is computed from A, because the paper does not give its own normalization. Both fits have the same error $J_H = 2434.42$, so they describe the same force.

Calibration

Drag force as the known reference (γ from PSD), $\mathbf{F}_{MT} \approx \gamma \, \dot{\mathbf{p}}$:

$$\textbf{Paper:} \qquad J_H(g_H, \hat{\mathbf{D}}_H) = \sum_j \left\| \gamma \, \dot{\mathbf{p}}(j) - g_H \, \mathbf{V}_H^{\mathrm{T}}(j) \, \hat{\mathbf{D}}_H^{\mathrm{T}} \mathbf{L}(\hat{\mathbf{p}}(j)) \, \hat{\mathbf{D}}_H \, \mathbf{V}_H(j) \right\|^2$$

$$\textbf{Dissertation:} \qquad J_H(\hat{f}_H, \hat{\mathbf{D}}_H, \mathbf{b}) = \sum_j \left\| \mathbf{F}_{MT}(j) - \hat{f}_H \, \hat{\mathbf{F}}_H\big(\mathbf{p}(j), \mathbf{b}, \mathbf{V}_H(j); \hat{\mathbf{D}}_H\big) \right\|^2$$

$$\boxed{J_H = 2434.42 \implies \sigma = \sqrt{J_H/3N} = 0.087 \text{ pN}} \qquad \text{(current-based: 0.179 pN)}$$

Result — two normalizations, same force

A. Paper (six gains free)

$g_H = 4.741 \text{ pN/V}^2$

$\hat{\mathbf{D}}_H = \mathrm{diag}(0.444, 0.364, 0.445, 0.436, 0.353, 0.436)$

B. Dissertation ($d_{H_1} \to 1$)

$\hat{f}_H = g_H (0.444)^2 \approx 0.93 \text{ pN/V}^2$

$\hat{\mathbf{D}}_H = \mathrm{diag}(1, 0.820, 1.002, 0.982, 0.795, 0.982)$

$$\text{bridge:} \qquad \hat{f}_H = g_H\, c^2, \qquad \hat{\mathbf{D}}_H^{(B)} = c^{-1}\hat{\mathbf{D}}_H^{(A)}, \qquad c = 0.444.$$

Sources: paper eq. (7),(10),(11),(17), pp. 2808–2812; dissertation eq. (4.2)–(4.4),(4.10), pp. 66–80.